Hackathon: Creating a Critical Persona FatBot

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대회방법

- 3인 1조로 토론 관련 대화 생성모델 대회 참전

  - 목표: 지금까지 배운 내용을 토대로 3시간 (두 강의) 동안 seq2seq 챗봇 모델을 강화하라!

  ~~- 상품: 고가의 미끼 제품들 준비~~

Rules

- Join the Debate-Themed Dialogue Generation Model Competition in Teams of Three
  - Goal: Strengthen your seq2seq chatbot model in 3 hours using all the knowledge you've gained so far!
  - ~~Prize: High-end bait items are prepared~~

Build your own Debater

- Create the world's most powerful "debate battle" language model.
- You may use any data you like. You can even create or generate your own data.
- You must use an LSTM-based seq2seq model as the base, but you may add or modify the model architecture.
  - (Optional) It would be great if the model could understand the entire context of previous debates and respond accordingly.

- Teams should be organized into three main roles: data preparation, model development, and strategy (writing a technical report and slide).
  - Once one part is finished, help each other dynamically. Grades are not affected by this, so just focus on it during class this week and next week!

Build your own Debater (Example of training data)


```
{
  "topic": "안락사 허용",
  "input_context": "고통스러운 삶을 강제로 이어가게 하는 것은 비인도적입니다.",
  "target_output": "하지만 생명의 가치는 타인이 판단할 수 없으며 남용의 소지가 큽니다.",

  "encoder_input": "안락사 허용 <SEP> 고통스러운 삶을 강제로 이어가게 하는 것은 비인도적입니다.",
  "decoder_input": "<SOS> 하지만 생명의 가치는 타인이 판단할 수 없으며 남용의 소지가 큽니다.",
  "decoder_target": "하지만 생명의 가치는 타인이 판단할 수 없으며 남용의 소지가 큽니다. <EOS>"
}
```

```
{
"topic": "Legalization of Euthanasia",
"input_context": "It is inhumane to force someone to continue a life full of suffering.",
"target_output": "However, the value of life cannot be judged by others, and there is a high risk of misuse.",

"encoder_input": "Legalization of Euthanasia <SEP> It is inhumane to force someone to continue a life full of suffering.",
"decoder_input": "<SOS> However, the value of life cannot be judged by others, and there is a high risk of misuse.",
"decoder_target": "However, the value of life cannot be judged by others, and there is a high risk of misuse. <EOS>"
}
```

Build your own Debater

- **An example of dataset for source**

https://huggingface.co/datasets/raphaaal/schopenhauer-debate

Search this dataset

history
string · *lengths*

547 758 40.4%

{'0': "<Opponent> Uninsured deposits had nothing to do with SVB's collapse. It was all about mismanagement. Stop blaming the depositors! </Opponent><You>…

{'1': "<Opponent> The Silicon Valley Bank collapse was inevitably caused by the rising interest rates and tough economic climate. Banks and regulators need to…

{'2': "<Opponent> Interest rates and economic climate? Really? The Silicon Valley Bank collapse was purely due to mismanagement and risky investments.…

{'3': "<Opponent> Silicon Valley Bank's management shouldn't be blamed for the bank's collapse. They were just caught in the crossfire of an unpredictable economic downturn. #SVB #NotTheirFault </Opponent><You> But isn't it the management's job to anticipate and navigate through economic uncertainties? They had risk management tools at their disposal. #SVB #AccountabilityMatters </You><Opponent> True, but the scale and speed of the downturn were unprecedented. It's easy to point fingers in hindsight, but they were dealing with circumstances beyond their control. #SVB #UnprecedentedTimes </Opponent>"}

"<Opponent> Silicon Valley Bank's management shouldn't be blamed for the bank's collapse. They were just caught in the crossfire of an unpredictable economic downturn. #SVB #NotTheirFault </Opponent>

<You> But isn't it the management's job to anticipate and navigate through economic uncertainties? They had risk management tools at their disposal. #SVB #AccountabilityMatters </You>

<Opponent> True, but the scale and speed of the downturn were unprecedented. It's easy to point fingers in hindsight, but they were dealing with circumstances beyond their control. #SVB #UnprecedentedTimes </Opponent>"

Tips

- The proposed dataset is too small making your model answer properly
  - I recommend you add discourse corpora to your model!! To understand the way of basic conversation for it
- More than 10K data are needed actually

- You may need a GPU (e.g., RTX 3080)

https://huggingface.co/datasets/mc-ai/conversation_dataset/viewer/default/train?row=2&views%5B%5D=train

https://huggingface.co/datasets/Isotonic/human_assistant_conversation

https://huggingface.co/datasets/SohamGhadge/casual-conversation

Thank you.